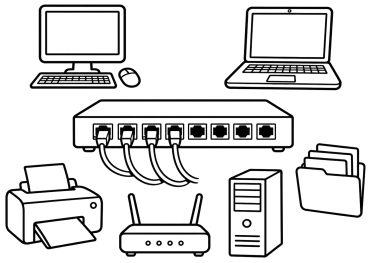
TEACHER GUIDE

ANSWERS AND MARKS

GRADE 9 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 5
Computer Networks



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 5 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - Activities 5.1 - 5.8, safety, textbook links and notes, a quiz
- **Answers** - chapters 1 to 3, in the same order as the workbook
- **Unit 5 Test** - the marking scheme and the common mistakes
- **Extra** - homework ideas, a practical test and teaching tips

★ BEFORE THE FIRST LESSON

- ☐ You know where the lab switch, the server and the router are.
- ☐ File and printer sharing work on the lab computers (Practicals 5 - 8).
- ☐ The msg command or a LAN messenger works in your lab (Practicals 3 - 4).

"Students understand a network best when they follow a real cable to a real switch."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Wired or wireless?	6
	B Name the media	6
	C Fill in the blanks	8
	D Complete Figure 5.1	7
	E True or false	6
	F Choose the correct answer	6
	G Answer these	4
Chapter 2	A Name the devices	6
	B Who am I?	5
	C Label the network	6
	D True or false	6
	E Draw a network	6
	F Crossword	10
	G Answer these	4
Chapter 3	A Complete the tree	6
	B Put the steps in order	5
	C What is shared?	8
	D Fill in the blanks	8
	E Match them	5
	F Spot the mistakes	5
	G Word search	10
	H Answer these	4
End	Unit 5 Test	40
TOTAL		**177**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Before the unit, find the lab network yourself: the switch, the server, the router and the cables that are safe to show.
- Collect the printouts of Practical 8 and show how one printer served the whole class.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (43)	Ch 2 (43)	Ch 3 (51)	Test (40)	Total (177)
1						
2						
3						
4						
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6						
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29						

TEACHER GUIDE

Practicals at a Glance

Practicals 1 to 8 = workbook activities 5.1 to 5.8

Practical	Students do	Check that ...
1 (5.1)	Find the devices of the lab network	they find the NIC port, the RJ45 plug, the switch, the server and the router; no cable is pulled out; the four lines are filled.
2 (5.2)	A network of two computers (Figure 5.5)	you plug in the cables; the switch lights come on; the other computer shows under Network.
3 (5.3)	A message with the CLI (Command Prompt)	they open cmd, type the msg command exactly and close the window with exit.
4 (5.4)	Messages with messenger software	one message to one friend and one message to many friends at one time; polite messages only.
5 (5.5)	Share a folder	the folder has a copy of two pictures and is shared with Everyone (Read).
6 (5.6)	Open a shared folder	they open a friend's folder from Network and copy one picture; nothing is deleted.
7 (5.7)	Share a printer (you show it)	they write the steps: Printers & scanners, Manage, Printer properties, Sharing.
8 (5.8)	Print on the network printer	one copy each; they choose the shared printer in the Print window.

! SAFETY AND MATERIALS

- Only you plug in or pull out network cables (Practicals 1 and 2). Keep cables off the floor.
- For Practical 2: two computers, a small switch and two RJ45 cables.
- Practicals 3 - 8: network discovery and file and printer sharing are on; use a teacher account.
- Practical 3: msg works on Windows 10 Pro; on Windows 10 Home use a LAN messenger (Practical 4).

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - What is a computer network?	66 - 68	-
Chapter 2 - Network devices	69 - 71	5.1, 5.2
Chapter 3 - Sharing and communication	71 - 73	5.3 - 5.8
Unit 5 in One Page	73 - 74 (summary)	-

★ TEACHING TIPS

- Bring old network parts to the class: an RJ45 cable, a network interface card, an old switch. Let students hold them.
- Let students make a poster of the lab network and keep it on the wall during the unit.

! TEXTBOOK NOTES - THE WORKBOOK IS CORRECT

- Page 73: "LAN messages" is the program **LAN Messenger**. "Net send" was a command of old Windows; Windows 10 Pro uses **msg** (msg * text = every user of the same computer; msg /server:NAME * text = another computer, when your network allows it).
- The textbook spells **fiber** (American) and writes "Wi-fi waves" in Figure 5.4. The workbook writes **fibre** and **Wi-Fi**. Accept both spellings.
- Page 72: "buying multiuser software licenses ... is relatively low" means the **cost** is low.
- The textbook has two sections: 5.1 (pages 66 - 71) and 5.2 (pages 71 - 73). The workbook numbers its sections 5.1 - 5.9 in its own order.
- The steps of Activities 5.1 - 5.8 are written from the textbook pages for Windows 10.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What is a computer network?	**Two or more computers connected to share resources**
2. Name the two types of network connections.	**Wired and wireless**
3. Name two wired media.	**Fibre optic and twisted pair cables**
4. Name two wireless media.	**Infrared, Bluetooth, Wi-Fi, radio waves**
5. Which device manages the files of a network?	**The server**
6. Which device is the mediator between computers?	**The switch**
7. Which card connects a computer to a network?	**The NIC**
8. Which device connects a network to the Internet?	**The router**
9. Name the two basic objectives of a network.	**Communication, sharing resources**
10. What does CLI stand for?	**Command Line Interface**

ANSWERS

Chapter 1

What is a computer network?

A Wired or wireless?

6 marks

The network connection	Wired	Wireless
Data travels through fibre optic cables.	☑	☐
Two mobile phones share a song with Bluetooth.	☐	☑
The lab computers are joined with twisted pair cables.	☑	☐
A laptop joins the school network with Wi-Fi.	☐	☑
Data is sent with infrared waves.	☐	☑
Data is sent with radio waves.	☐	☑

B Name the media

6 marks

1. Wi-Fi waves	**2.** Fibre optic cable	**3.** Bluetooth
4. Radio waves	**5.** Twisted pair cable	**6.** Infrared waves

1 mark each. Accept "Wi-Fi" for 1 and "fiber optic cable" for 2.

C Fill in the blanks

8 marks

1. network	**2.** two	**3.** resources	**4.** wireless
5. twisted pair	**6.** stand-alone	**7.** radio	**8.** hard

D Complete Figure 5.1

7 marks

1. users	**2.** photos	**3.** games	**4.** storage
5. server	**6.** hardware	**7.** Internet	

1 mark each (the seven advantages of Figure 5.1, textbook page 67).

E True or false

6 marks

1. F	**2. T**	**3. F**	**4. T**	**5. T**	**6. F**

False: 1 (a network has two or more computers), 3 (Bluetooth is wireless), 6 (a network gives access to the Internet).

F Choose the correct answer

6 marks

1. (b) network	**2.** (c) Wi-Fi waves	**3.** (d) Fibre optic
4. (a) server	**5.** (b) hard copy	**6.** (b) two

G Answer these

4 marks

1. Name the two types of network connections. Give one example of each. (2 marks)

Wired, e.g. fibre optic cables or twisted pair cables (1 mark). Wireless, e.g. infrared waves, Bluetooth, Wi-Fi or radio waves (1 mark).

2. Your school lab has 20 computers and one printer. Write two advantages of joining them in a network. (2 marks)

Any two (1 mark each): every computer can print on the one printer (sharing hardware); common use of documents, photos, songs and videos; sharing software such as games; e-mail and short messages; access to the Internet; easy back ups on the server.

ANSWERS

Chapter 2

Network devices

A Name the devices

6 marks

1. Switch	**2.** Router	**3.** Server
4. RJ45 cable	**5.** Network Interface Card (NIC)	**6.** Fibre optic cable

1 mark each. Accept "network card" for the NIC and "network cable" for the RJ45 cable.

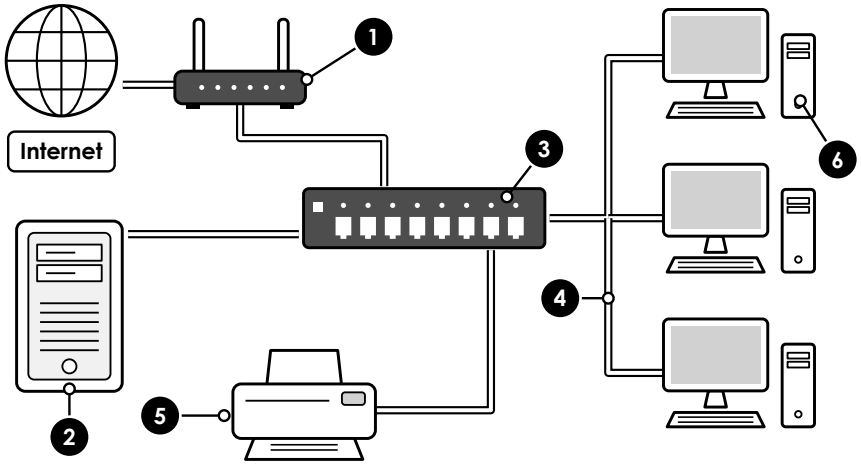
B Who am I?

5 marks

1. Server	**2.** Switch	**3.** Network Interface Card (NIC)
4. Router	**5.** RJ45 cable	

C Label the network

6 marks



1. **Router**
2. **Server**
3. **Switch**
4. **RJ45 cable**
5. **Printer**
6. **Computer**

D True or false

6 marks

1. T	**2. F**	**3. T**	**4. F**	**5. T**	**6. F**

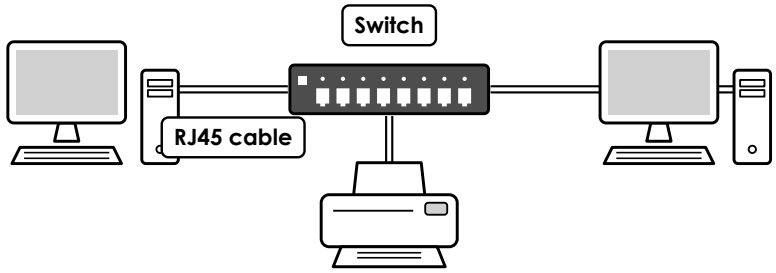
False: 2 (the switch recognises the destination and sends the message there), 4 (a network may have different servers), 6 (they are joined through the switch).

TEACHING TIP

- Read the crossword clues aloud for weaker readers. All the words are in the word list.

E Draw a network

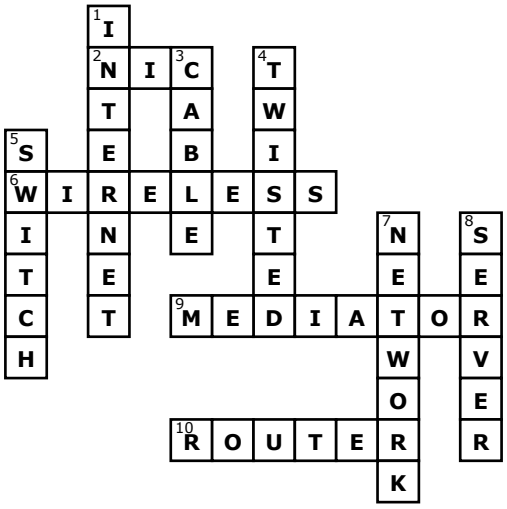
6 marks



Two computers (1 mark), a printer (1), a switch (1), a cable from each device to the switch (2), the labels Switch and RJ45 cable (1).

F Crossword

10 marks



Across: 2 NIC, 6 WIRELESS, 9 MEDIATOR, 10 ROUTER

Down: 1 INTERNET, 3 CABLE, 4 TWISTED, 5 SWITCH, 7 NETWORK, 8 SERVER

G Answer these

4 marks

1. Nimal gives a print command on Computer 2. How does the command reach the printer? (2 marks)

It goes through the switch (1 mark). The switch recognises the destination device, the printer, and sends the command to it (1 mark).

2. What is the difference between a switch and a router? (2 marks)

A switch is the mediator between the computers of a network: it sends each message to its destination (1 mark). A router connects one network to another network or to the Internet (1 mark).

★ TEACHING TIPS

- Draw Figure 5.6 on the board step by step: the server, the switch, then each computer with its own cable.
- Say it every lesson: the switch is the mediator inside the network; the router is the door to other networks and the Internet.
- Act out a switch: one student is the switch and passes paper messages only to the right "computer".

ANSWERS

Chapter 3

Sharing and communication

A Complete the tree

6 marks

1. Communication	**2.** Sharing resources	**3.** Messaging
4. E-mailing	**5.** Software and hardware	**6.** Data and information

1 mark each. Boxes 3 and 4 may be in either order, and so may boxes 5 and 6.

B Put the steps in order

5 marks

The numbers in the boxes, from the top down (1 mark each):

1. 4	**2. 1**	**3. 5**	**4. 2**	**5. 3**

C What is shared?

8 marks

In the network ...	Communi-cation	Hardware	Software	Data and information
All the lab computers print on one printer.	☐	☑	☐	☐
The school buys one multiuser licence.	☐	☐	☑	☐
Sports meet photos are in a shared folder.	☐	☐	☐	☑
Sunil sends one message to all his friends.	☑	☐	☐	☐
All the office computers use one scanner.	☐	☑	☐	☐
Students play a game shared on the network.	☐	☐	☑	☐
The teacher e-mails homework to the class.	☑	☐	☐	☐
Concert songs are in a shared folder.	☐	☐	☐	☑

D Fill in the blanks

8 marks

1. time	**2.** typed	**3.** Graphical	**4.** licence
5. money	**6.** storage	**7.** folder	**8.** fax

E Match them

5 marks

1. CLI → **C**	**2.** Multiuser licence → **E**	**3.** Shared folder → **A**
4. Peripherals → **D**	**5.** LAN Messenger → **B**	

TIP: Let pairs swap books and check each other's tick tables before you mark them.

F Spot the mistakes

5 marks

1. selling → **sharing**
2. only one friend at a time → **many friends at one time**
3. with pictures and icons → **typed as text**
4. only one computer can use it → **everyone in the network can use it**
5. wastes → **saves**

1 mark for each mistake that is found and corrected.

G Word search

10 marks

	H									
	A									
	R	S	O	F	T	W	A	R	E	
P	D			M						
R	W	L	I	C	E	N	C	E		
I	A					S		L		F
N	R	E	N	N	A	C	S	I		O
T	E							A		L
E								M	G	D
R	S	H	A	R	I	N	G	E		E
				A	T	A	D			R

Word	Starts at	Direction
SHARING	row 10, col 2	across
PRINTER	row 4, col 1	down
SCANNER	row 7, col 8	across, backwards
FOLDER	row 6, col 11	down
MESSAGE	row 4, col 5	slanting down
EMAIL	row 10, col 9	up
SOFTWARE	row 3, col 3	across
HARDWARE	row 1, col 2	down
LICENCE	row 5, col 3	across
DATA	row 11, col 8	across, backwards

H Answer these

4 marks

1. **How does sharing software in a network save money? (2 marks)**
   Each stand-alone computer may need its own software licence, which is very expensive (1 mark). A multiuser licence for the network costs much less (1 mark).

2. **What is a shared folder? Name two things friends can share with it. (2 marks)**
   A folder shared in a network: all the files in it can be used by the network users (1 mark). Any two: images, photos, songs, videos, documents (1 mark).

★ TEACHING TIPS

- Use the opener story: who has carried a pen drive to the printer computer? A network solves that problem.
- Show the Wi-Fi and Bluetooth signs on a phone. Ask which media are wired and which are wireless.
- Before Practical 3, test the msg command on two lab computers. On Windows Home, use a LAN messenger instead.

MARKING SCHEME

Unit 5 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (d) Keyboard	**2.** (b) Bluetooth
3. (a) NIC	**4.** (a) switch
5. (b) Router	**6.** (b) sharing
7. (b) CLI	**8.** (b) A and C only

2 Fill in the blanks

8 marks (1 each)

1. twisted	**2.** radio	**3.** server	**4.** cables
5. e-mailing	**6.** fax	**7.** storage	**8.** Line

3 Look at the pictures

8 marks (1 + 1 each)

1. Switch - the mediator: it sends each message to the right computer or device	**2. Router** - connects a network to another network or to the Internet
3. Network Interface Card (NIC) - connects a computer to a computer network	**4. Bluetooth** - a wireless medium (a wireless connection)

1 mark for the name and 1 mark for what it does or what type it is.

4 Answer these

16 marks

1. What is a computer network? (2 marks)

A collection of two or more computers connected together (1 mark) to share resources (1 mark).

2. Write three advantages of computer networking. (3 marks)

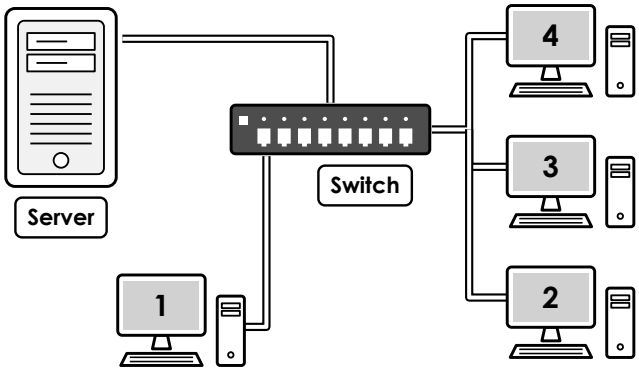
Any three (1 mark each): communicate by e-mail and short messages; common use of documents, images, photos, songs, videos; share software such as games; save information in a network storage device; easy back ups on the server; share software and hardware; access the Internet.

3. Name the two types of network connections. Give two examples of the media of each type. (3 marks)

Wired and wireless (1 mark). Wired: fibre optic cables, twisted pair cables (1 mark). Wireless: any two of infrared waves, Bluetooth, Wi-Fi, radio waves (1 mark).

4. Draw a network of three computers connected to a server through a switch. Label the server and the switch. (3 marks)

Three computers and a server (1 mark), each joined to the switch by its own cable (1 mark), the labels Server and Switch (1 mark).



5. What are the two basic objectives of a computer network? (2 marks)

Communication (1 mark) and sharing resources (1 mark).

6. Write three advantages of using a printer in common in a network. (3 marks)

Several computers can use common hardware (1 mark); it saves money (1 mark); it saves time (1 mark).

WHAT STUDENTS USUALLY GET WRONG

- They say a network can be one computer. A network has **two or more** computers.
- They mix up the **switch** (the mediator inside one network) and the **router** (joins networks and the Internet).
- They think the switch sends every message to all the computers. It finds the **destination**.
- They think a server must be a special machine. **Any ordinary computer** with the necessary software can be a server.
- They call Bluetooth and Wi-Fi cables. They are **wireless** media.
- They write that a multiuser licence costs more. It costs **less** than a licence for every computer.
- They think the CLI uses pictures and icons. It uses **typed commands** as text.

TEACHING TIPS

- Before Practical 5, turn on network discovery and file and printer sharing on the lab computers.
- Teach students to share a copy, not the only copy, and never to delete files from a friend's folder.
- In Practical 7, share the printer once on the teacher computer. Students watch and write the steps.

HOMEWORK AND EXTRA PRACTICE

- Draw the network of your home or of a shop you know. Which devices are wired, and which are wireless?
- List three things your class can share in the lab network: one hardware, one software, one data.
- Write one sentence for each basic device of a network: what does it do?
- Fast learners: count the ports of the lab switch. How many more computers could it join?

SET-UP NOTES FOR THE PRACTICALS (WINDOWS 10)

- **Computer name** (needed in Practicals 3 and 6): Settings → System → About → Device name.
- **Network discovery and file sharing**: Control Panel → Network and Sharing Center → Change advanced sharing settings: Turn on network discovery, Turn on file and printer sharing.
- **msg * Hello** shows the message to every user of the same computer.
  msg /server:NAME * Hello sends it to the computer NAME, when the network allows it. Windows 10 Home has no msg.
- A free LAN messenger program on every computer is the easiest way to send one message to the whole lab.

PRACTICAL TEST - 20 MINUTES, 20 MARKS (OPTIONAL)

In the lab, one student at a time, after the unit. Read the steps aloud. Mark on the screen.

Step	The student must ...	Marks
1	Point to the NIC port and the RJ45 cable of a computer and name them.	2
2	Show the switch of the lab and say what it does.	2
3	Say which computer is the server and one thing it manages.	2
4	Open the Command Prompt and type the message command you give.	3
5	Make a folder, put one picture in it and share it with Everyone.	4
6	Open a friend's shared folder from Network and copy one file.	3
7	Print a one-page letter on the shared printer.	2
8	Close every window and log off.	2
	Total	**20**

TEACHING TIPS

- Talk about money: compare thirty single software licences with one multiuser licence for a lab of thirty computers.
- Use the word list as a quick spelling test at the start of a lesson.